

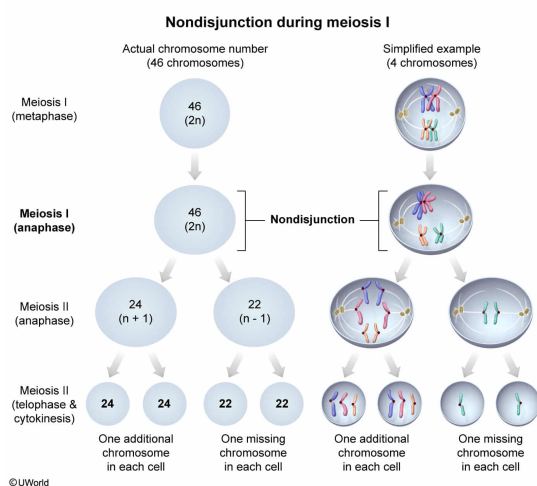
A single diploid cell containing 46 chromosomes underwent meiosis and produced four daughter cells, two with 22 chromosomes and two with 24 chromosomes. An error in which of the following stages of meiosis would most likely explain this result?

- ☐ A. Metaphase II [11%]
☐ B. Prophase II [4%]
☐ C. Telophase I [4%]
✔ ☒ D. Anaphase I [79%]

Correct

79% answered correctly

Explanation:



In eukaryotic organisms, gametes (ie, eggs, sperm) form via **meiosis**, a process of cellular division resulting in the production of four haploid (n) daughter cells from a single diploid ($2n$) parental cell. Meiosis occurs in two stages, meiosis I and meiosis II:

- **Meiosis I** begins with prophase I, in which the nuclear envelope of a diploid parental cell disintegrates and homologous chromosomes (each consisting of two sister chromatids) pair up. In metaphase I, these chromosome pairs align at the metaphase plate. The chromosome pairs are separated to opposite poles of the cell by the meiotic spindle in anaphase I. The nuclear envelope then reforms in telophase I and the cytoplasm of the cell divides (cytokinesis), yielding two haploid daughter cells.
- In **Meiosis II**, each daughter cell undergoes a process similar to meiosis I, except that it is *sister chromatids*, not homologous chromosomes, that are pulled to opposite poles of the cell by the meiotic spindle. After each cell divides, the result is a total of four haploid daughter cells.

Normally, when a diploid cell with 46 chromosomes undergoes meiosis, the four resulting daughter cells each normally contain 23 chromosomes. However, in the given scenario two daughter cells have an *extra* chromosome (24 total) and two are *missing* a chromosome (22

total). Such a finding is most likely due to **chromosomal nondisjunction**, which occurs when either homologous chromosomes (during anaphase I) or sister chromatids (during anaphase II) **fail to separate**.

Chromosomal nondisjunction during **anaphase I** commonly produces one daughter cell with an **extra homologous chromosome** (24 chromosomes) and one daughter cell **missing a chromosome** (22 chromosomes) at the end of meiosis I. Further division of these cells in meiosis II (*without* nondisjunction) would then yield a total of four cells, two with 22 chromosomes each and two with 24 chromosomes each.

(Choices A, B, and C) Errors in anaphase—*not* prophase, metaphase, or telophase—result in nondisjunction and unequal numbers of chromosomes in daughter cells. In addition, errors in **meiosis II** typically result in two normal daughter cells and two daughter cells with differing abnormal chromosome counts.

Educational objective:

During meiosis, chromosomal nondisjunction occurs when homologous chromosomes (in meiosis I) or sister chromatids (in meiosis II) fail to separate to opposite poles of the cell during anaphase. These events lead to extra chromosomes in some daughter cells and missing chromosomes in others.

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